

ENSC 225 – Lab #3: Characterization of BJTs and single stage BJT amplifier

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Abstract

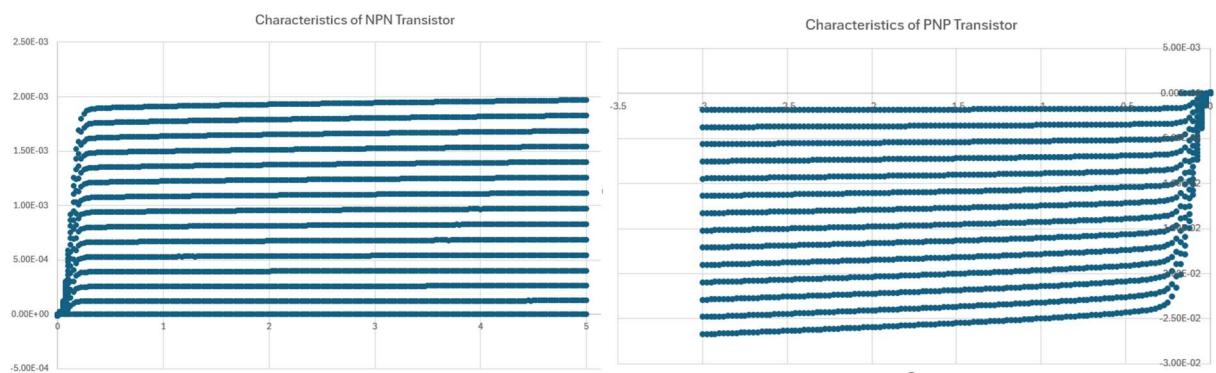
This laboratory experiment focuses on characterizing NPN and PNP bipolar junction transistors (BJTs) and analyzing their performance in amplifier circuits. Using the CA3046 NPN-BJT array and the 2907 PNP transistor, students measure key parameters such as the IC-VCE characteristics, forward current gain (β), and Early voltage. The large-signal behavior of the NPN transistor is also explored through input-output voltage plots.

The second part of the lab involves designing and testing a single-stage common-emitter amplifier with a DC collector current of 500 μA . Small-signal voltage gain and frequency response are measured and compared with LT-Spice simulations. The experiment provides practical experience with transistor characterization, amplifier design, and performance analysis in both the time and frequency domains.

Experiment 1,2 & 3:

Introduction: In the first section of the lab, we successfully characterized the NPN transistor (Q1) from the CA3046 chip by plotting its IC vs. VCE characteristics using the Source Measurement Unit (SMU) system. The data was collected over 15 sweeps, with current and voltage values accurately recorded. Similarly, in the second section, we characterized the PNP 2907 transistor by sweeping VCE from 0 to -3V and IB from 0 to -10 μA , successfully obtaining the IC-VCE characteristics for the PNP transistor as well. Both sets of data were successfully acquired, providing the necessary measurements for further analysis of the transistors' electrical properties.

Results:



	NPN	PNP
DC β	$\beta(500\mu A) = \frac{I_C}{I_B} = \frac{0.0005025}{4 \times 10^{-6}} = 125.625$ $\beta(1mA) = \frac{0.0010111}{8 \times 10^{-6}} = 126.3875$ $\beta(2mA) = \frac{0.002034}{16 \times 10^{-6}} = 127.125$	$\beta(5mA) = \frac{0.005025}{2 \times 10^{-3}} = 251.25$ $\beta(10mA) = \frac{0.010106}{-2 \times 10^{-5}} = 252.65$
Early Voltage	$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{0.000545 - 0.00053}{4.85 - 1} = 3.8691 \times 10^{-6}$ $b = y - mx$ $b = 0.000533 - (3.8691 \times 10^{-6})(1.5) = 5.27 \times 10^{-4}$ $y=0 \text{ find } x$ $x = \frac{y-b}{m} = \frac{0 - 5.27196 \times 10^{-4}}{3.8691 \times 10^{-6}} = -136.258V$ $V_A = -136.26V$	$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{(-0.00743) - (-0.00706)}{(-2.5) - (-0.95)} = 2.3871 \times 10^{-4}$ $b = y - mx$ $b = (-0.00721) - (2.3871 \times 10^{-4})(-1.5) = -6.85193 \times 10^{-3}$ $y=0 \text{ find } x$ $x = \frac{y-b}{m} = \frac{0 - (-6.85193 \times 10^{-3})}{2.3871 \times 10^{-4}} = 28.70399$ $V_A = 28.70V$

Comparing NPN values with data sheet

V_{CE}	I_C	Data Sheet β	Graphical β	Data sheet V_A	Graphical V_A
3V	500 μA	100	125.625	61.1	136.26V
3V	1 mA	100	126.3875	61.1	136.26V
3V	2mA	100	127.125	61.1	136.26V

Comparing PNP values with data sheet

V_{CE}	I_C	Data Sheet β	Graphical β	Data sheet V_A	Graphical V_A
2V	5 mA	100	251.25	44.3	28.70V
2V	10 mA	100	252.65	44.3	28.70V

Discussion:

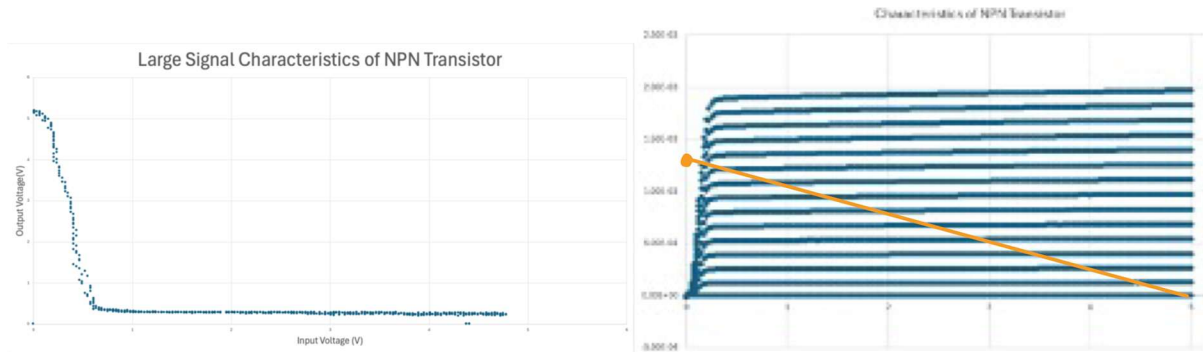
Our NPN Beta is close to expected. The difference in the Early voltage V_A values could be due to variations in measurement conditions, such as temperature, biasing, sweep inaccuracies, and collector current. The datasheet value of 61.1 is typically for standard test conditions, while the measured value of 136 may reflect specific conditions in our experiment or differences between individual transistors. For the PNP are slightly under the expected data sheet but the beta voltages are again off, this could be for the same reasons as mentioned above.

Experiment 4:

Introduction: In this section, we analyze the large-signal behavior of the NPN transistor by applying a varying input signal and measuring the resulting output voltage. This allows us to plot the transistor's large-signal transfer characteristics, which reveal how it responds to non-linear input voltages. Additionally, we overlay the load line on the I_C vs. V_{CE} characteristics to compare experimental and theoretical results. This analysis helps us understand the transistor's behavior in real-world applications, particularly in amplifier circuits.

Resistors Used:

	12k Ω	2k Ω
Measured Value	11.184	2.167



Load Line Calculations:

$$V_{CE} = V_{CC} - I_C R_C$$

$$I_C = \frac{V_{CC}}{R_C} = \frac{5V}{2.167k} = 2.307mA$$

	Load Line
V_{CE}	5V
I_C	2.307mA

Discussion:

In this experiment, we applied a varying input signal to the NPN transistor and measured the corresponding output voltage to plot the large-signal transfer characteristics. The load line, determined using the collector resistor value, was drawn on the I_C vs. V_{CE} characteristics. The experimental data showed the transistor's non-linear behavior, consistent with the expected saturation and cutoff regions. Any discrepancies between the experimental and theoretical results could be attributed to factors such as component tolerances, measurement inaccuracies, or approximations in the load line calculation.

Experiment 5:

Introduction: In this section, we design and analyze a single-stage common-emitter amplifier using an NPN transistor. The goal is to bias the transistor for a DC collector current of 500 μA and a collector voltage of 3V, using appropriate resistor values for the base and collector to achieve the desired operating point. We then apply a small signal to the base, measure the output, and calculate the small-signal voltage gain. Additionally, the lab explores the amplifier's frequency response and performance under different operating conditions. This exercise helps us understand the practical aspects of amplifier design, including biasing, gain calculation, and how BJTs are used in real-world amplification applications.

	Component Value	Measured Value
R_{S1}	1 k Ω	0.987 k Ω
R_{S2}	100 k Ω	98.61 k Ω
R_B	500 k Ω	464 k Ω
R_C	12 k Ω	11.806 k Ω
R_E	12 k Ω	11.846 k Ω
C_{C1}	1 μF	1.003 μF

Part A – Measured DC supply voltages and nodes voltages

Voltage	Measured Value
V_{CC}	9.061V
V_{EE}	-9.055 V
V_B	-2.871 V
V_C	3.759 V
V_E	-3.663 V

Part B – Determining the Transistor Currents

I_B	I_C	I_E
$I_B = \frac{V_B - 0}{R_B}$ $I_B = \frac{-2.871}{464}$ $I_B = -6.188 \times 10^{-6} \text{ A}$	$I_C = \frac{V_{CC} - V_C}{R_C}$ $I_C = \frac{9.061 - 3.759}{11.806}$ $I_C = 4.491 \times 10^{-4} \text{ A}$	$I_E = \frac{V_{EE} - V_E}{R_E}$ $I_E = \frac{-9.055 + 3.663}{11.846}$ $I_E = 4.552 \times 10^{-4} \text{ A}$

Part C – Determining the DC β of the transistor

Comparison:

Calculations to find β :

$$\beta = \left| \frac{I_C}{I_B} \right|$$

$$\beta = \left| \frac{4.491 \times 10^{-4}}{-6.188 \times 10^{-6}} \right|$$

$$\beta = 72.576$$

The value for β found in the SPA was around 126, which is higher than the value found after our calculations, 72.576. The SPA measurement reflects the transistor's actual performance, and the jump between the measurement can be attributed to differences between idealized calculations and real-world conditions

Part D&F – Oscilloscope Screen Captures (Non By-pass and By-pass)

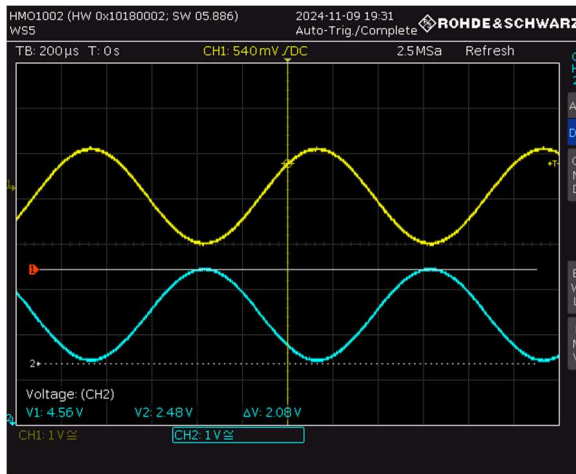


Figure 1: Non by-pass emitter screen capture

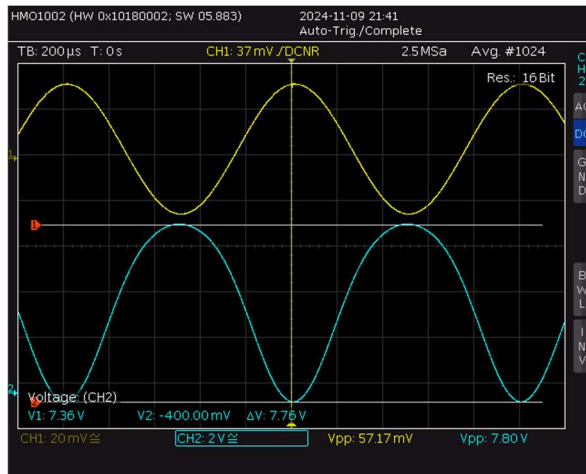


Figure 2: By-passed emitter screen capture

Part E&G – Computing Gains for Small Signal Voltage and Estimating Gains

	Input	Output
Non By-pass	2.12V	2.08V
By-pass	20.43mV	2.95V

Non By-pass

$$\text{Small Signal Gain: } \frac{V_C}{V_{Sig}} = \frac{V_{Output}}{V_{Input}} = \frac{2.08}{2.95} = 0.9813$$

By-pass

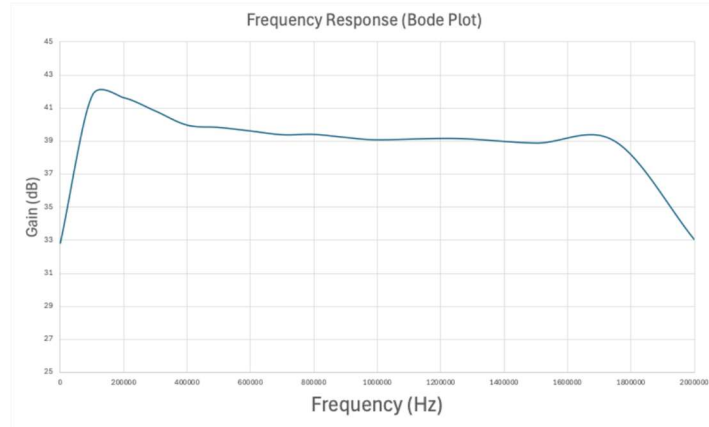
$$\text{Small Signal Gain: } \frac{V_C}{V_{Sig}} = \frac{V_{Output}}{V_{Input}} = \frac{2.95}{0.02043} = 144.395$$

$$\text{Estimated Gain: } \frac{V_C}{V_B} = \frac{3.759}{2.871} = 1.309$$

Part H – Max Pk-Pk Without Distortion

We measured **11.84V** before the signal became distorted.

Part I – Frequency Response (Blode Plot)



Part J&K – DC Operating Points and Frequency Response

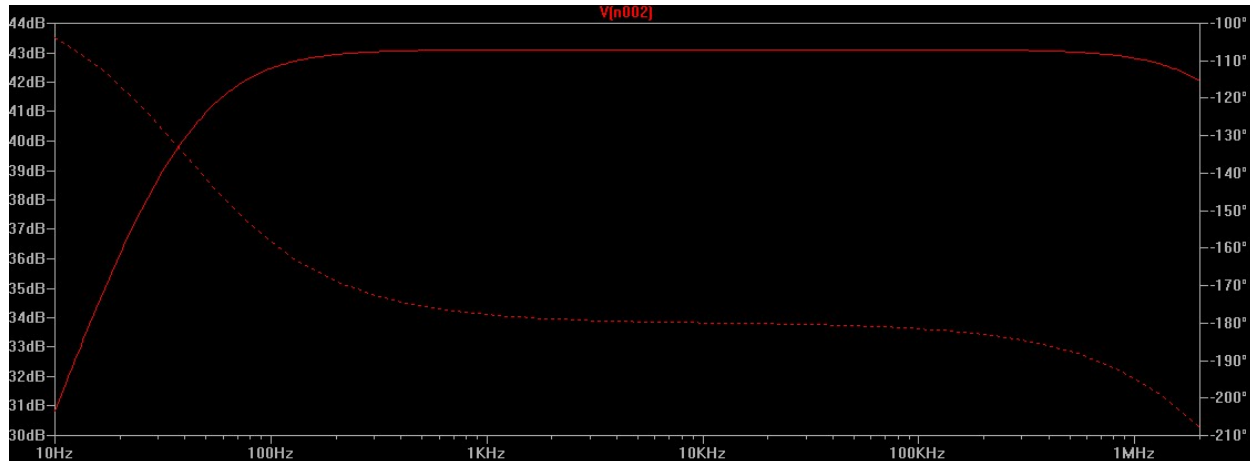
DC Operating Points (LTSpice Simulation)

V_B	V_C	V_E
-3.418 V	4.278 V	-4.169

V_B	V_C	V_E
-2.871 V	3.759 V	-3.663 V

The LTSpice DC values are all slightly larger by around 0.5V.

LTSpice Frequency Response Plot



Part L&N – Mid-band Gain Comparison and Signal Distortion

Midband Gain with By-pass Capacitor

Theoretical	Simulated	Experimental
$A_V = 20 \log \left(\left -\frac{\beta R_C}{r_n} \right \right)$ $= 20 \log \left(\left -\frac{72.576 \times 11.806 \times 10^3}{4151.58} \right \right)$ $= 46.29 \text{ dB}$	43.2 dB	39.63 dB

$$r_n = \left| \frac{V_T}{I_B} \right| = \frac{25.69 \times 10^{-3}}{6.188 \times 10^{-6}} = 4151.58 \Omega$$

Discussion

While increasing the amplitude slowly, we observed the output signal showing signal distortion first at 0.05V. When simulating inside of LTSpice, we also observed the exact same thing at 0.05V. If the bias is too low, the transistor enters the cutoff region, causing distortion at lower input levels. If the bias is too high, the transistor enters saturation, leading to clipping. In our circuit, distortion occurred at an output voltage of 0.05V, indicating that the transistor could no longer linearly amplify the signal. When comparing this to the LT-Spice simulation, the distortion behavior was similar, suggesting that the biasing values were appropriately set in both the real and simulated circuits. Adjusting the biasing could extend the linear range and help reduce early distortion.

Conclusion

In this experiment, we characterized both NPN and PNP transistors by examining their IC vs VCE characteristics, forward current gain (β), and Early voltage. Using the SPA, we were able to plot the characteristics of the NPN (CA3046) and PNP (2907) transistors, providing a detailed understanding of their behavior under varying base currents and collector-emitter voltages.

From the IC vs VCE characteristics of the NPN transistor, we determined the DC current gain (β) and Early voltage at different collector currents (500 μ A, 1 mA, and 2 mA) at VCE = 3V. Similarly, for the PNP transistor, we calculated β and Early voltage at different collector currents (5 mA and 10 mA) at VEC = 2V. These experimental results were compared with the datasheet values and showed reasonable agreement, with some discrepancies likely due to real-world variations such as temperature effects and measurement limitations.

In addition, the large signal characteristics of the NPN transistor were explored by applying a slow varying input signal and plotting the output vs input voltage. From this data, we constructed the large signal transfer characteristics and drew the load line on the NPN IC vs VCE characteristics. This allowed us to compare the experimental data with theoretical predictions. The analysis of any discrepancies between the experimental and theoretical results provided valuable insights into the transistor's non-linear behavior under large signal conditions.

Lastly, Part 5 successfully demonstrated the design, analysis, and testing of a single-stage common-emitter amplifier. By carefully selecting the biasing resistors and components, we were able to achieve the desired DC operating point and verify the amplifier's performance both experimentally and through simulation. We observed that the amplifier exhibited distortion when the input amplitude exceeded the transistor's linear operating range, and we successfully measured and analyzed the frequency response. The LT-Spice simulations closely matched the experimental results, providing confidence in our design and biasing choices. By bypassing the emitter resistor, we were able to achieve higher gain at mid frequencies, confirming the importance of feedback in amplifier design. This lab reinforced the fundamental principles of transistor amplifier design and provided valuable insights into the behavior of common-emitter amplifiers in real-world circuits.